ANTI-BIOFILM POTENTIAL OF ONION PEEL EXTRACT AGAINST *Bacillus subtilis*

A PROJECT REPORT

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BONAFIDE CERTIFICATE

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INTERNAL EXAMINER

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ABSTRACT

The formation of biofilm serves as a vital survival mechanism for numerous bacterial species, such as *Bacillus subtilis*, allowing them to attach to surfaces and withstand the effects of antibiotics, disinfectants, and the host's immune system. Biofilms cause persistent infections, industrial clogging, and corrosion, resulting in substantial public health and economic problems. *B. subtilis*, a non-pathogenic Gram-positive bacterium, is used as a model organism for studying biofilm development due to its complex regulatory pathways and extracellular matrix production. The escalation of antimicrobial resistance has heightened the need for alternative methods to counter biofilms. Products from nature, particularly those made from waste generated by agricultural industries, have been recognized for their sustainable and bioactive properties. A discarded onion skin is a valuable source of phenolic compounds and flavonoids with well-documented antimicrobial and antioxidant characteristics. An investigation into the anti-biofilm capabilities of onion peel extract was conducted against *B. subtilis*. The extract was processed using solvent extraction techniques and its effectiveness was assessed through biofilm inhibition tests, microscopic analysis, and quantification methods. The researchers examined the effects of disruption on the biofilm structure, with particular emphasis on how it interacts with quorum sensing and matrix formation. Research indicates that the onion peel extract demonstrates notable anti-biofilm properties, underscoring its capability as a natural, environmentally friendly substitute for traditional treatments. This study implies the biofilm control by employing plant-based substances and promotes the value of agricultural waste, which is in line with the goals for sustainable health and environmental management.

Keywords: Biofilm, *Bacillus subtilis*, Antimicrobial resistance, Onion peel extract, Quorum sensing, Agricultural waste, Sustainable treatment.

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LIST OF SYMBOLS AND ABBREVIATIONS

%	PERECNTAGE
g	GRAM
mg	MILLIGRAM
µg	MICROGRAM
cm	CENTIMETER
mm	MILLIMETER
µm	MICROMETER
nm	NANOMETER
µM	MICROMOLAR
µl	MICROLITRE
ml	MILLILITRE
α	ALPHA
℃	DEGREE CELSIUS
v/v	VOLUME BY VOLUME
w/v	WEIGHT/VOLUME
Dia	DIAMETER
Rad	RADIUS
Avg	AVERAGE
Rpm	REVOLUTIONS PER MINUTE
dil.	DILUTE

Conc.	CONCENTRATION
GAE	GALLIC ACID EQUIVALENTS
QE	QUERCETIN EQUIVALENTS
TPC	TOTAL PHENOLIC CONTENT
TFC	TOTAL FLAVONOID CONTENT
NaOH	SODIUM HYDROXIDE
NaCl	SODIUM CHLORIDE
MnCL2	MANGANESE CHLORIDE
HCl	HYDROCHLORIC ACID
FeCl3	FERRIC CHLORIDE
NB	NUTRIENT BROTH
CFU	COLONY FORMING UNIT
PBS	PHOSPHATE BUFFERED SALINE
EPS	EXTRACELLULAR POLYMERIC SUBSTANCES
TNF	TUMOR NECROSIS FACTOR
IL	INTERLEUKIN
DNA	DEOXY RIBONUCLEIC ACID
LDL	LOW DENSITY LIPOPROTEIN
ROS	REACTIVE OXYGEN SPECIES

CHAPTER 1

INTRODUCTION

1.1 BIOFILMS

Microbial communities form complex structures encased in a self-made matrix of extracellular polymeric substances (EPS), which attach to both living and non-living surfaces. These formations serve as a survival mechanism for bacteria, allowing them to withstand various environmental stresses, such as antimicrobial agents, oxidative stress, nutrient deficiencies, and the immune responses of the host. The development of biofilms plays a pivotal role in the long-term persistence and chronic nature of bacterial infections. Biofilms that contain bacterial cells can be 1000 times more resistant to antibiotics than cells that exist independently in a free-living state.

The escalating worldwide threat of antimicrobial resistance has highlighted the need to develop innovative approaches to tackling biofilm-related infections. Antibiotics commonly used in traditional medicine frequently fail to eliminate bacteria embedded in biofilms, which causes persistent infections, treatment failures, and greater healthcare burdens. Exploiting natural compounds with anti-biofilm properties offers a sustainable and promising approach for investigation.

1.2 IMPORTANCE OF BIOFILMS

Biofilms are more than just groups of bacteria; they consist of intricate, cooperative structures that encompass sophisticated methods of communication among cells (quorum sensing), mutual resource sharing, and coordinated defensive strategies. The EPS matrix, consisting of polysaccharides, proteins, nucleic acids, and lipids, functions as a physical barrier and biochemical defence that shields the bacteria from antibiotics, disinfectants, and immune reactions.

Microbial communities can be found in a variety of environments, including dental plaque and catheter surfaces within the human body, as well as wastewater pipelines, industrial cooling systems, and marine vessels. Biofilms are a considerable issue in various environments due to their widespread presence and ability to persist.

Biofilms are responsible for approximately 80% of all recurring infections within the healthcare system, encompassing periodontitis, persistent wounds, urinary tract infections, and conditions arising from implanted medical equipment such as catheters, pacemakers, prosthetics, and heart valves. Biofilm control has become a crucial objective in contemporary microbiology and the study of infectious diseases due to the extensive participation of various elements.

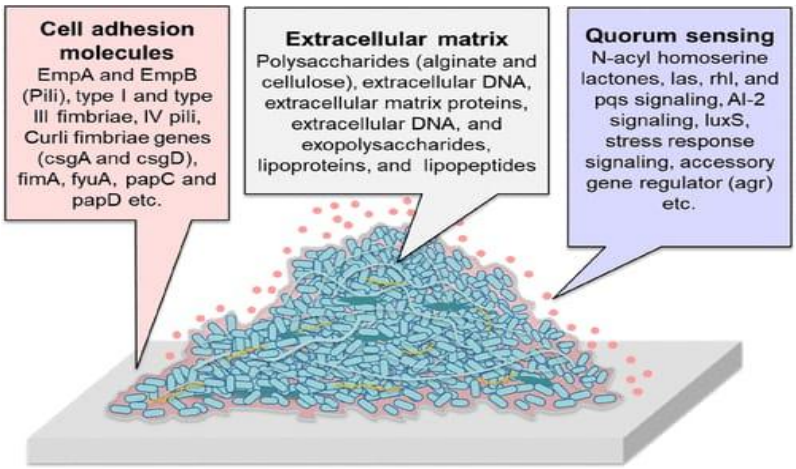


Fig 1.1 Components of Biofilm

1.3 BIOFILM-ASSOCIATED INFECTIONS

Research on biofilms typically focuses on well-known disease-causing bacteria such as *Pseudomonas aeruginosa, Staphylococcus aureus, or Escherichia coli*, but non-pathogenic organisms like *Bacillus subtilis* also have a significant impact in biofilm studies. Bacillus subtilis is a Gram-positive, rod-shaped bacterium which is commonly found in soil and the human and animal gastrointestinal tract. It is commonly regarded as harmless, and is in fact utilised as a probiotic in fermented foods, while also serving well as a model organism in biofilm research.

The biofilm architecture of *B. subtilis* is well-documented and comprises multiple genetic pathways, such as those controlling the synthesis of EPS(Extracellular Polymeric Substances), amyloid-like fibres (TasA), and surfactin. The spo0A, sinR, and eps operons are crucial in controlling the formation of biofilms in these species. The intricate regulatory processes involved make *B. subtilis* a strong model organism for studying the formation of biofilms in Gram-positive bacteria and evaluating possible anti-biofilm treatments.

B. subtilis biofilms are generally not linked to human infections but do have practical applications in industry. Clogs and contamination can be caused by them in bioreactors or fermentation equipment. Examining its biofilms offers insight into the fundamental principles of biofilm biology and the effectiveness of anti-biofilm treatments.

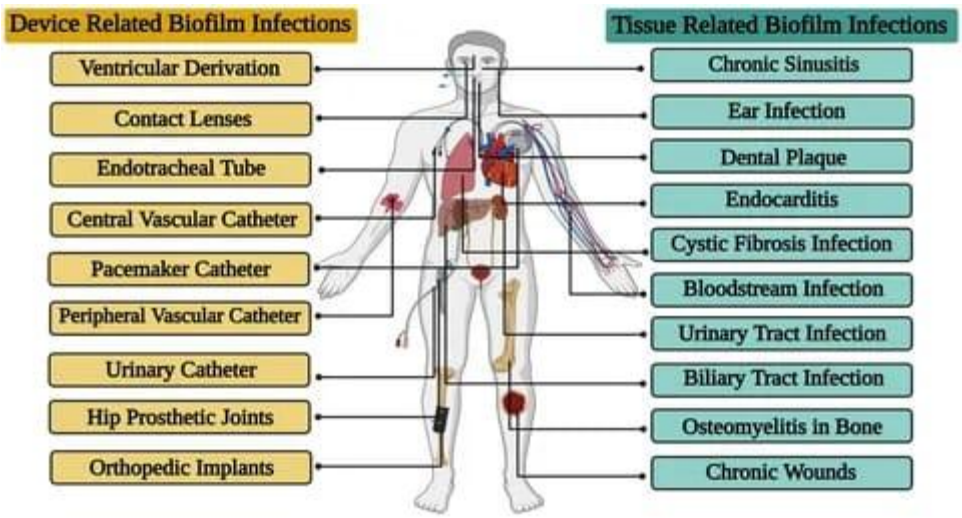


Fig 1.2 Biofilm Based Infections

1.4 NATURAL ANTI-BIOFILM AGENTS

In response to the constraints of conventional antibiotics, researchers have begun to focus on compounds derived from nature, with a particular emphasis on those possessing antimicrobial, antioxidant, and anti-inflammatory properties. Flavonoids, alkaloids, terpenoids, and polyphenols - phytochemicals found in

plants - have demonstrated potential in preventing the formation of biofilms and breaking up established biofilms.

They exert their effects through several mechanisms. They are as follows; Prevention of bacterial attachment to surfaces, Interfering with the quorum sensing mechanisms, Disrupting the manufacturing process for energy performance standards, Bacterial cells are subjected to oxidative stress, Modifying the permeability of bacterial membranes.Fruit, vegetable, and plant waste abundant flavonoids have garnered substantial interest because of their wide-ranging effects and relatively low toxicity levels.

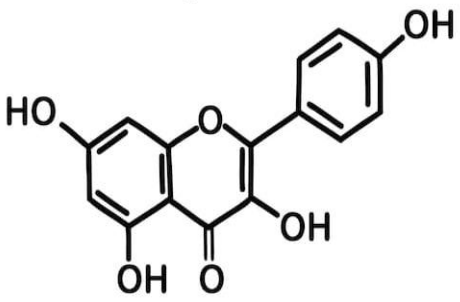


Fig1. 3 Kaempferol

1.5 SIGNIFICANCE OF ONION PEEL AS A WASTE-DERIVED COMPOUND

The global food industry produces significant quantities of agricultural waste, with a substantial portion comprising the peels and skins of vegetables such as onions. A discarded onion peel is a rich source of bioactive compounds, specifically quercetin, a flavonoid. Utilizing such waste is consistent with the goals of sustainable development, achieved by encouraging waste valorization, protecting the environment, and enhancing resource efficiency.

Findings from recent studies have indicated that onion peels possess considerably greater quercetin concentrations than the edible bulb. They offer a cost-effective

and environmentally friendly means of extracting valuable bioactive compounds with possible therapeutic and industrial uses.

Experiments have shown that extracts from onion peels, dissolved in methanol or ethanol, exhibit antimicrobial, antioxidant, and anti-inflammatory properties. Research is currently being conducted to explore the potential applications of certain agents in controlling biofilms, with early findings suggesting they can prevent biofilm formation and disintegrate existing biofilms through several mechanisms.

1.6 QUERCETIN: CONTENT, PROPERTIES, AND ANTI-BIOFILM POTENTIAL

A naturally occurring flavonol known as Quercetin, with the chemical formula 3,3′,4′,5,7-pentahydroxyflavone, is found in a variety of plant-based sources such as fruits, vegetables, and plant residues. Notable sources include apples, grapes, capers, berries, and particularly onion skins. The Quercetin content in onion peels can reach as much as 1,600 mg/100g in their dry weight, depending on factors such as plant type, maturity, and the environment, making them a valuable resource.

Pharmacological properties of Quercetin are Antioxidants work by neutralizing free radicals and subsequently decreasing oxidative stress, This compound has anti-inflammatory properties, which involve blocking the action of inflammatory mediators like TNF-α and IL-6,Interferes with bacterial membrane integrity, halts DNA gyrase function, and reduces enzyme activity levels, This compound has anti-cancer properties, affecting the signalling pathways of cancer cells and triggering programmed cell death in these cells, Protects the heart and brain by reducing oxidation of bad cholesterol and enhancing the functioning of blood vessel linings, Cardio protective and Neuro protective: Reduces LDL oxidation and improves endothelial function.

Anti-Biofilm Activity of Quercetin are Quercetin displays significant anti-biofilm activity via several distinct pathways, The inhibition of quorum sensing molecules, including acyl-homoserine lactones, in Gram-negative bacteria, Regulation of biofilm-related genes is decreased, specifically lasR, rhlR, and algD,Decreases in EPS production and disruption to established biofilm structures, Prevention of cell-cell attachment and initial surface settlement.

In *Bacillus subtilis*, quercetin has been found to disrupt the regulatory pathways linked to matrix production and quorum sensing, resulting in decreased biofilm formation and structural instability. These results confirm its potential as a non-toxic, plant-based biofilm inhibitor.

1.7 APPLICATIONS OF BIOFILM INHIBITION

Biofilm inhibition has significant implications across multiple fields, particularly in medical, agricultural, industrial, and environmental domains. The ability of plant-derived extracts, such as onion peel extract, to inhibit biofilm formation presents eco-friendly and sustainable alternatives to synthetic antimicrobials.

1.7.1 MEDICAL APPLICATIONS

Biofilms are a major concern in clinical settings due to their role in chronic infections and antibiotic resistance. Inhibiting biofilm formation:Prevents colonization on medical devices (e.g., catheters, implants, stents),Enhances antibiotic efficacy against resistant bacterial strains.

1.7.2 FOOD AND AGRICULTURE

In food processing and agriculture, biofilms contribute to contamination and spoilage: Prevents biofilm development on food-contact surfaces and packaging, reduces bacterial persistence on crop surfaces or in irrigation systems, Natural

antimicrobials from food waste (e.g., onion peel) offer cost-effective solutions for crop protection and food hygiene.

1.7.3 INDUSTRIAL AND ENVIRONMENTAL SECTORS

Biofilms cause fouling in pipelines, reactors, and filtration systems: Biofilm inhibitors can improve water treatment, reduce corrosion, and increase equipment lifespan, environmentally friendly agents help manage biofilms without harsh chemicals.

1.7.4 PHARMACEUTICAL AND BIOTECHNOLOGICAL RESEARCH

Plant-based antibiofilm compounds are being explored for new drug development, also serve as leads for quorum-sensing inhibitors and biofilm-dispersing agents.

1.8 OBJECTIVES OF THE STUDY

1. To extract and quantify the bioactive flavonoid quercetin from onion peels using solvent extraction methods.

2. To assess the anti-biofilm activity of onion peel extract against Bacillus subtilis using standard in vitro assays.

3. To evaluate the structural and functional changes in the biofilm matrix after treatment with the extract using microscopic and quantitative methods.

4. To promote the use of agricultural waste for developing eco-friendly antimicrobial agents that align with sustainable health and environmental goals.

CHAPTER 2

REVIEW OF LITERATURE

2.1 BIOFILM FORMATION AND STAGES

Communities of microorganisms are formed in a structured manner, attached to a surface, and encased in a self-generated extracellular matrix consisting of polysaccharides, proteins, and nucleic acids. Biofilm formation progresses through a series of distinct phases such as; Loosely attached planktonic bacteria utilise vander Waals forces and fimbriae to adhere to a surface initially., Bacterial cells become more securely attached through the use of extracellular polymeric substances (EPS), Maturation stages I and II involve the thickening of the biofilm, resulting in a complex three-dimensional structure with water channels that facilitate nutrient flow, Cells or groups of cells separate and return to a floating state, allowing them to disperse and colonize new surfaces.

The progression of these stages is carefully controlled by gene expression and environmental signals such as nutrient availability, pH, and quorum sensing mechanisms (Hall-Stoodley *et al.,* 2004).

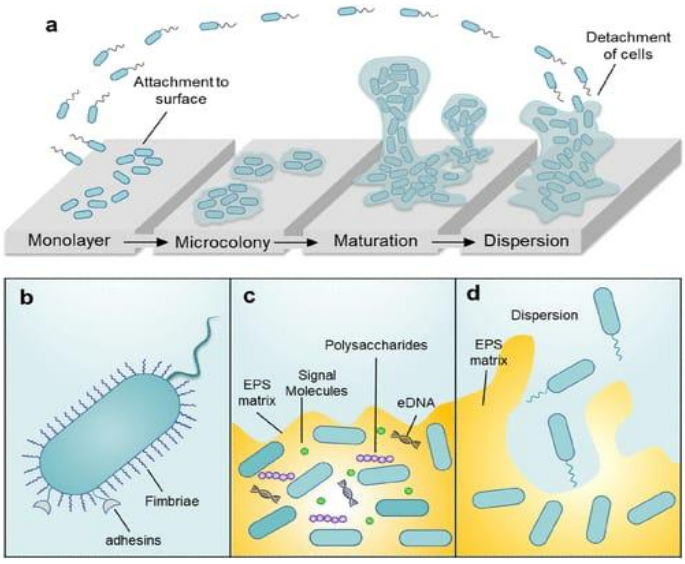


Fig 2.1 Biofilm Formation Stages.

2.2 MECHANISM OF RESISTANCE IN BIOFILMS

A major worry about biofilms is their ability to withstand antimicrobial treatments. Planktonic bacteria show significantly less resistance compared to bacteria found in biofilms, with the latter's resistance being as high as 1,000 times greater. The processes involved in resistance encompass.

The EPS matrix restricts the penetration of antibiotics. The altered microenvironment leads to gradients in oxygen and nutrients, causing metabolic variation, which in turn results in cells entering a dormant state and becoming less susceptible to antibiotic effects. Efflux pumps are upregulated in biofilms to expel antimicrobial agents. Survival under hostile conditions is enhanced by Stress Response Genes. Quorum sensing facilitates a coordinated response to antimicrobial agents. This complex resistance is a significant reason for ongoing and repeated infections.

2.3 ANTIBIOFILM AGENTS FROM NATURAL COMPOUNDS

Traditionally restricted antibiotics have limited effectiveness in treating biofilm-related infections, prompting researchers to investigate natural substances, particularly those found in plants, for their potential to prevent biofilm formation. Compounds of natural origin act to target different stages of biofilm developmentas follows; Initial adhesion is prevented by substances such as tannins and saponins. Quorum sensing can be disrupted by compounds such as furanones and eugenol. Prevention of EPS synthesis (e.g., flavonoids such as quercetin).The dispersal of biofilm and the destruction of non-reproductive cells. Plants produce various secondary metabolites, including alkaloids, terpenoids, phenolics, and flavonoids, which exhibit considerable antibiofilm properties (Borges *et al.,* 2016).

2.4 ANTIMICROBIAL COMPOUNDS IN ONION PEEL (QUERCETIN, FLAVONOIDS)

Quercetin, specifically the compound 3,3′,4′,5,7-pentahydroxyflavone, is: Quercetin (3,3′,4′,5,7-pentahydroxyflavone) is: An effective antioxidant that has been proven to eliminate reactive oxygen species (ROS).A substance that reduces inflammation by regulating cytokine function. This compound has antimicrobial properties that enable it to break down bacterial membranes and suppress the activity of critical bacterial enzymes.

Onion peels are a source of additional phenolic compounds including kaempferol, myricetin, and rutin, which contribute to their antibacterial and antifungal effects. Waste onion peel has been shown to contain 1,161 to 1,625 mg of quercetin per 100g dry weight, which is significantly higher than that in the edible bulb (Zaborowski *et al*., 2024).

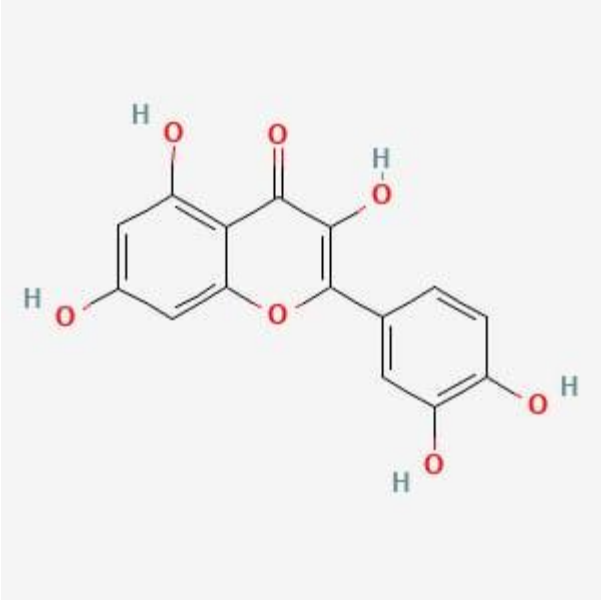


Fig 2.2 Quercetin

Quercetin compromises the structural integrity of biofilms through its actions. Preventing the production or activity of quorum sensing molecules, such as autoinducer-2. Inhibiting the production of EPS synthesis and biofilm structural

proteins. Making bacterial cells more responsive to antibiotics involves weakening their outer membrane defenses. Onion peel possesses certain characteristics that make it a promising, eco-friendly natural source of antibacterial and antibiofilm agents.

2.5 *Bacillus subtilis* BIOFILM MECHANISMS

The non-pathogenic *Bacillus subtilis* is a Gram-positive soil bacterium frequently used as a model organism in studies of biofilm formation. The biofilm structure is composed of the following: The TasA amyloid fibers serve as a structural framework. Exopolysaccharides (EPS) play a key role in cell attachment and the process of colonizing surfaces. The BslA protein creates a hydrophobic protective layer on the biofilm's surface.

Biofilm formation in *B. subtilis* is controlled by multiple signalling pathways. Phosphorylation of Spo0A triggers the expression of biofilm-related genes. The SinR/SinI pathway functions as a primary controller of the activation and repression of matrix genes. The ComX quorum-sensing peptide facilitates communication between cells. *B. subtilis* biofilms replicate the structure and behavior of pathogenic biofilms, making them a useful model for studying antibiofilm interventions in a safe and controlled manner.

CHAPTER 3

MATERIALS AND METHODS

3.1 CHEMICALS AND REAGENTS

Nutrient broth, Nutrient Agar, Agar powder, Gallic acid, Quercetin, Ferric chloride, HCl, NaoH, Ethanol, Crystal violet solution, Distilled water.

3.2 COLLECTION AND PREPARATION OF ONION PEEL

Red onion (Allium cepa) peels were collected from local markets and thoroughly washed with tap water followed by distilled water to remove dirt and impurities. The cleaned peels were shade-dried at room temperature for 7–10 days until completely dehydrated. Dried peels were ground into a fine powder using a mechanical grinder and stored in airtight containers at room temperature.

3.3 EXTRACTION USING ETHANOL

About 25 g of powdered onion peel was soaked in 250 ml of 70% ethanol in a conical flask, sealed with parafilm, and kept on a rotary shaker at 120 rpm for 48 hours at room temperature. After extraction, the mixture was filtered using Whatman No.1 filter paper. The filtrate was evaporated to dryness under reduced pressure using a rotary evaporator or allowed to air-dry in a clean, dust-free environment. The dried extract was collected, weighed, and stored at 4°C for further analysis.

3.4 PHYTOCHEMICAL SCREENING

3.4.1 TEST FOR PHENOLS (FERRIC CHLORIDE TEST)

Procedure: 1 ml of ethanol extract was mixed with 3–5 drops of 1% ferric chloride solution.

$$TPC\ (mg\ GAE/g) = (C \times V) \div M$$

The above formula was used to calculate the Total Phenolic Content (TPC).

3.4.2 TEST FOR FLAVONOIDS (ALKALINE REAGENT TEST)

Procedure: 1 ml of extract was treated with a few drops of 10% sodium hydroxide solution. Few drops of dil. HCl were added.

$$TFC\ (mg\ QE/g) = (C \times V) \div M$$

The above formula was used to calculate the Total Flavonoid Content (TFC).

3.5 ANTIBACTERIAL ACTIVITY AGAINST *BACILLUS SUBTILIS*

The ethanolic extract of onion peel was evaluated for antibacterial activity against *Bacillus subtilis* using the agar well diffusion method.

Two wells were prepared on a Mueller-Hinton Agar plate:

Well 1 (Solvent Control): 50 µl of ethanol (used to dissolve the extract)

Well 2 (Extract): 50 µl of onion peel extract

3.6 BACTERIAL STRAIN AND CULTURE CONDITIONS

Bacillus subtilis (laboratory strain) was used in this study. Culture was maintained on nutrient agar slants at 4°C and sub-cultured routinely before experimental use.

For all experiments, overnight cultures were prepared by inoculating a single colony into 5 ml of nutrient broth and incubating at 37°C for 16–18 hours under shaking conditions (150 rpm).

3.7 PREPARATION OF SUPPLEMENTED NUTRIENT BROTH FOR BIOFILM FORMATION

To enhance biofilm formation, nutrient broth (NB) was supplemented with biofilm-inducing components. The composition for 250 ml of medium was as follows:

Table 3.1 Base: Nutrient Broth (250ml)

Component	Quantity
Peptone	1.25g
Beef Extract	0.75g
NaCl	1.25g
Distilled water	250ml

Table 3.2 Biofilm –Enhancing Supplements (250ml):

Supplement	Final concentration	Quantity
Glycerol	0.5% (v/v)	1.25ml
L-Glutamic acid	0.5% (w/v)	1.25g
$MnCl_2$(Manganese Chloride)	50µM	6.27mg
$FeCl_3$(Ferric Chloride)	50µM	8.11mg

The base medium (NB + glycerol + glutamic acid) was autoclaved at 121°C for 15 minutes. $MnCl_2$ and $FeCl_3$ were prepared as sterile stock solutions, filter sterilized using 0.22 µm syringe filters, and added aseptically after the medium cooled to ~50°C.

3.8 COLONY BIOFILM ASSAY

Solid medium was prepared by adding 1.5% agar to the supplemented NB. Plates were poured with 25 ml of medium and allowed to solidify. A 2 µl aliquot of overnight culture was spotted in the centre of each plate and incubated statically at 30°C for 72 hours. Colony morphology was visually examined and photographed. The presence of wrinkles, thickness, and surface structure were used to assess biofilm formation.

3.9 PELLICLE BIOFILM ASSAY (TEST TUBE METHOD)

Static biofilm formation at the air–liquid interface was assessed using sterile glass test tubes containing 5 ml of supplemented NB. Tubes were inoculated with 1% overnight culture and incubated statically at 30°C for 48–72 hours. Pellicle formation was observed visually and scored based on thickness and integrity.

3.10 MICROTITER PLATE BIOFILM ASSAY (96 WELL PLATE)

Biofilm biomass was quantified using the crystal violet staining method:

1. Inoculation: 200 µl of supplemented NB was added to each well of a sterile 96-well microtiter plate, followed by 2 µl of overnight culture.
2. Incubation: Plates were incubated statically at 30°C for 48 hours.
3. Washing: Wells were gently washed three times with sterile PBS to remove non-adherent cells.
4. Staining: 200 µl of 0.1% crystal violet was added and incubated for 15 minutes.
5. Destaining: Wells were rinsed again and dried, followed by the addition of 200 µl of 30% acetic acid to solubilize the bound dye.
6. Quantification: The absorbance was measured at 590 nm using a microplate reader.

Each experiment was performed in triplicate, and the average optical density (OD590) was calculated to estimate the relative biofilm biomass. The biofilm inhibition percentage was calculated by the below formula;

$$\text{Biofilm Inhibition (\%)} = ((OD_{CONTROL} - OD_{SAMPLE}) \div OD_{CONTROL}) \times 100$$

3.11 BIOFILM VISUALIZATION USING LIGHT MICROSCOPY

1. Preparation and Inoculation: Sterile glass coverslips were aseptically placed in each sterile petri plate. Each well was filled with 3 ml of Nutrient Broth (NB) supplemented with 1% glucose, $MnCl_2$, $FeCl_3$.Test compounds were added to treatment plates at predetermined concentrations to evaluate biofilm inhibition. Controls included a positive control (bacteria without inhibitor) and a negative control (medium without bacteria).
2. Incubation: The petri plates were incubated under static conditions at 37°C for 24–48 hours to allow for biofilm formation on the surface of the coverslips.
3. Washing: Following incubation, the coverslips were carefully removed using sterile forceps and gently rinsed twice with sterile phosphate-buffered saline (PBS) to remove non-adherent (planktonic) cells without disturbing the attached biofilm.
4. Staining: The washed coverslips were immersed in 0.1% (w/v) crystal violet solution for 15 minutes at room temperature. Excess stain was removed by gently rinsing the coverslips with sterile distilled water. The stained coverslips were allowed to air dry completely.
5. Microscopic Examination: Dried coverslips were mounted onto clean glass microscope slides. The stained biofilms were observed under a light microscope using 10× and 40× objectives. Photomicrographs were captured for documentation, and biofilm formation was qualitatively assessed based on cell aggregation and surface coverage.

CHAPTER 4

RESULTS AND DISCUSSIONS

4.1 PREPARATION OF ONION PEEL

Red onion (Allium cepa) peels were collected as agro-waste, washed with tap and distilled water, and shade-dried at room temperature for 7 days. The dried peels were ground into fine powder using a mechanical grinder and stored.

Fig 4.1 Onion peel powder

4.2 PREPARATION OF INOCULUM

A pure culture of Bacillus subtilis was inoculated into 10 ml of sterile nutrient broth and incubated at 37°C for 18–24 hours. After the observation of growth in nutrient broth, the bacteria were inoculated in the biofilm enhancing supplement nutrient media.

Fig 4.2 Streak plate

4.3 PREPARATION OF PRODUCTION MEDIA

The production medium was prepared using nutrient broth supplemented with biofilm-enhancing agents. For 250 ml of medium, 6.5 g of nutrient broth powder was dissolved in distilled water. Supplements such as glucose (1%), calcium chloride (0.1%), $FeCl_3$,$MnCl_2$ were added as required. The medium was autoclaved at 121°C for 15 minutes and cooled to room temperature before inoculation.

Fig 4.3 Biofilm Formation

4.4 PREPARATION OF EXTRACT SOLUTION

The dried onion peel extract was dissolved in ethanol to prepare a working solution of 100 µg/ml by dissolving 1 mg of extract in 10 ml ethanol. The solution was stored at 4°C for further analysis.

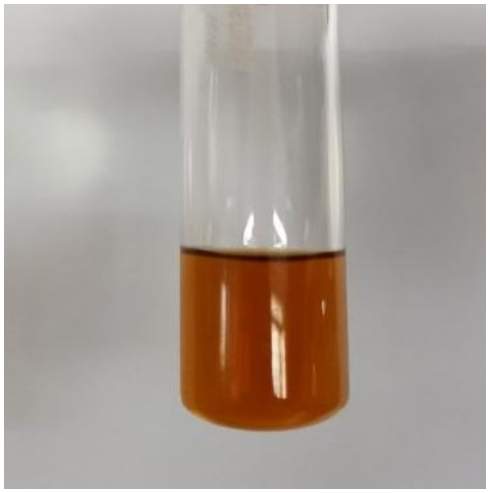


Fig 4.4 Onion peel (Ethanolic Extract)

4.5 PHYTOCHEMICAL SCREENING

4.5.1 TEST FOR PHENOLS (FERRIC CHLORIDE TEST)

1 ml of the onion peel ethanol extract was treated with a few drops of 1% ferric chloride solution. Formation of a deep green color indicated the presence of phenolic compounds.

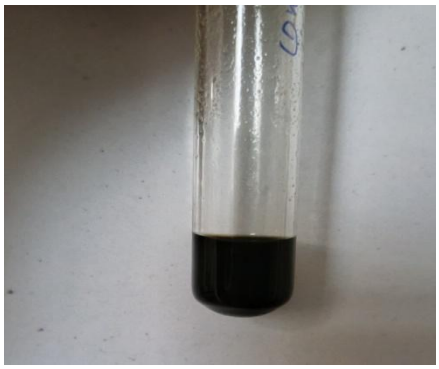


Fig 4.5 Phenol Test- Sample + Ferric Chloride

Phenol reacts with Ferric Chloride gives deep green colour. It confirms the presence of Phenolic compound in the sample.

4.5.2 TEST FOR FLAVONOIDS (ALKALINE REAGENT TEST)

1 ml of the extract was mixed with 1 ml of 10% sodium hydroxide solution. A yellow colour appeared, which turned colourless after adding a few drops of dilute hydrochloric acid, confirming the presence of flavonoids.

Fig 4.6 Sample+ extract Fig 4.7 Add of NaOH Fig 4.8 Add of dil. HCl

After adding NaOH to Sample yellow colour was formed. Then add dil. HCl to it, yellow colour disappears. It confirms the presence of Flavonoid in the sample.

4.6 ESTIMATION OF TOTAL PHENOLIC CONTENT (TPC)

Total phenolic content was estimated using the ferric chloride reagent method. To 1 ml of onion peel extract, 1 ml of 1% ferric chloride solution was added and mixed. The formation of a green or blue color was measured spectrophotometrically at 760 nm. Results were expressed as mg gallic acid equivalents (GAE) per gram of extract based on a standard calibration curve.

Table 4.1 Estimation of TPC

CONCENTRATION(µg/ml)	ABSORBANCE @760nm
10	0.15
20	0.27
40	0.51
60	0.75
80	0.99
100	1.23

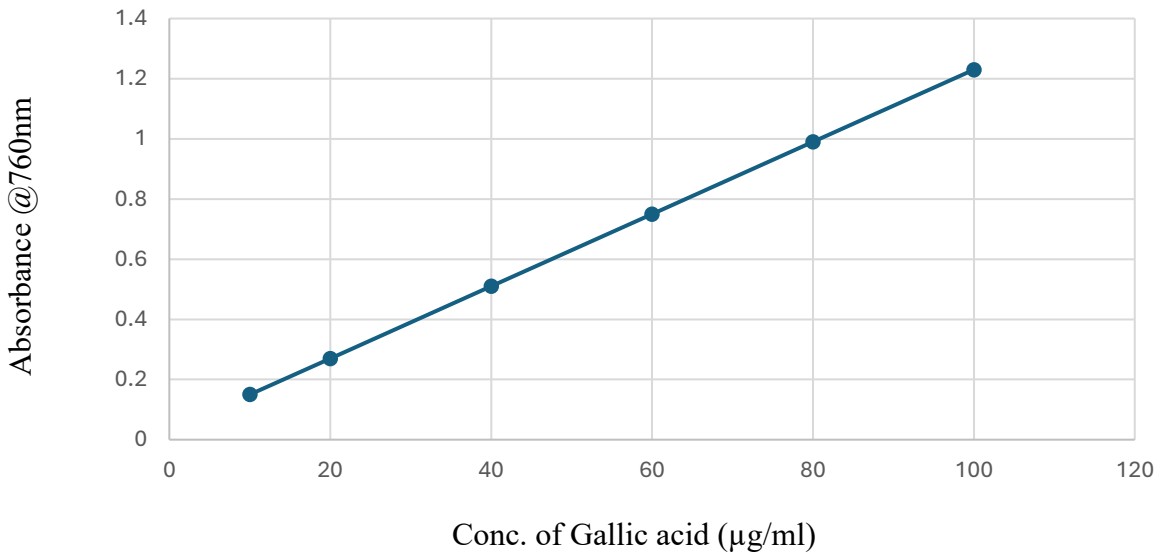


Fig 4.9 Standard Graph for TPC

4.7 ESTIMATION OF TOTAL FLAVONOID CONTENT (TFC)

Total flavonoid content was determined by treating 1 ml of the extract with 1 ml of dilute hydrochloric acid (HCl). After incubation, the change in colour was measured at 420 nm using a spectrophotometer. Quercetin was used as the

standard, and flavonoid content was expressed as mg Quercetin equivalents (QE) per gram of extract.

Table 4.2 Estimation of TFC

CONCENTRATION(µg/ml)	ABSORBANCE @420nm
10	0.115
20	0.207
40	0.385
60	0.565
80	0.745
100	0.925

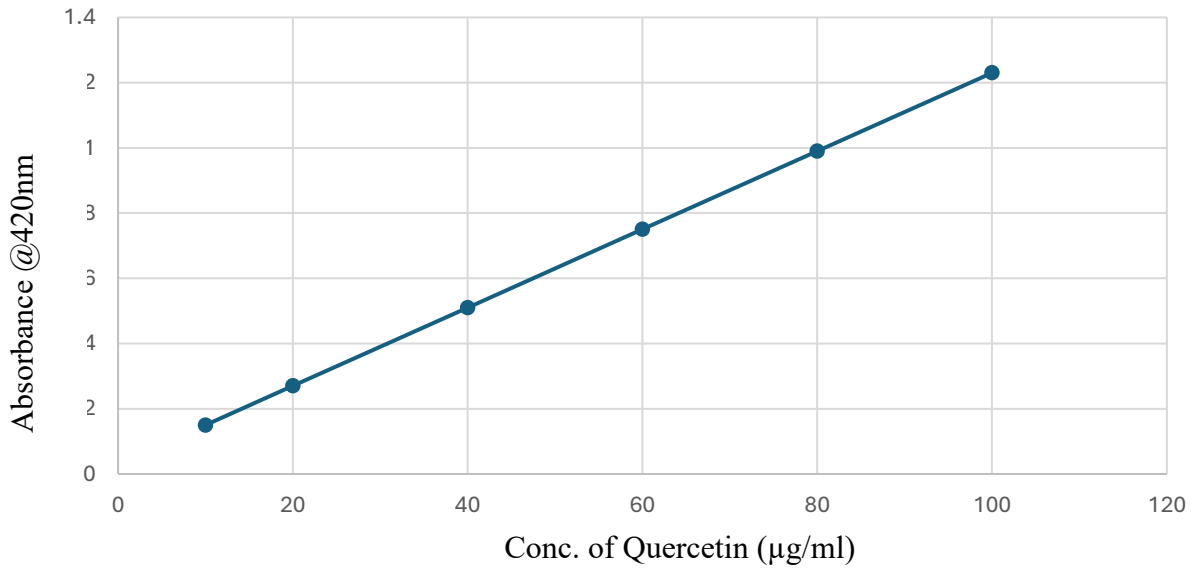


Fig 4.10 Standard Graph of TFC

Table 4.3 Estimation of TPC and TFC contents

SAMPLE	TOTAL PHENOLIC CONTENT (mg GAE/g)	TOTAL FLAVONOID CONTENT (mg QE/g)
Onion peel (Ethanolic Extract)	98.5	65.2

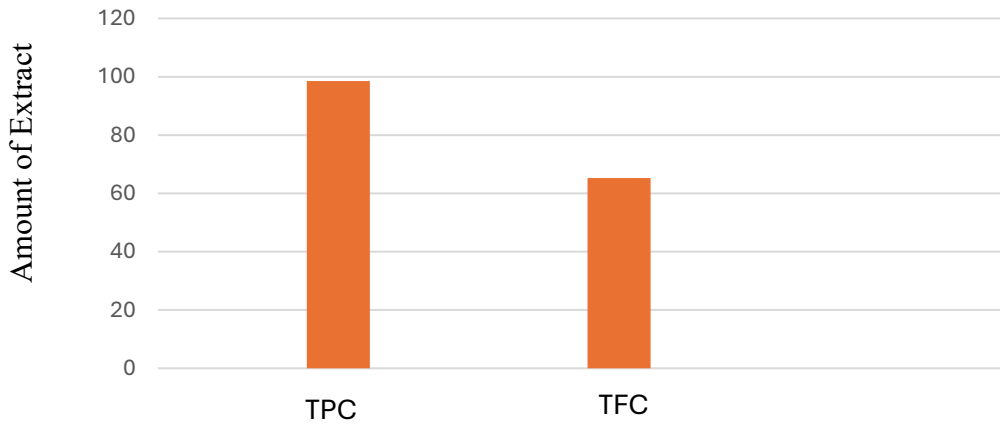


Fig 4.11 Bar Graph of TPC & TFC

4.8 ANTIBACTERIAL ASSAY

An antibacterial assay using the agar well diffusion method was conducted to test onion peel extract against a bacterial strain (*Bacillus subtilis)*. Mueller-Hinton Agar plates were inoculated with the test bacteria, and wells were filled with the extract and ethanol (as control). After incubation at 37°C for 24 hours, zones of inhibition were observed to assess antibacterial activity.

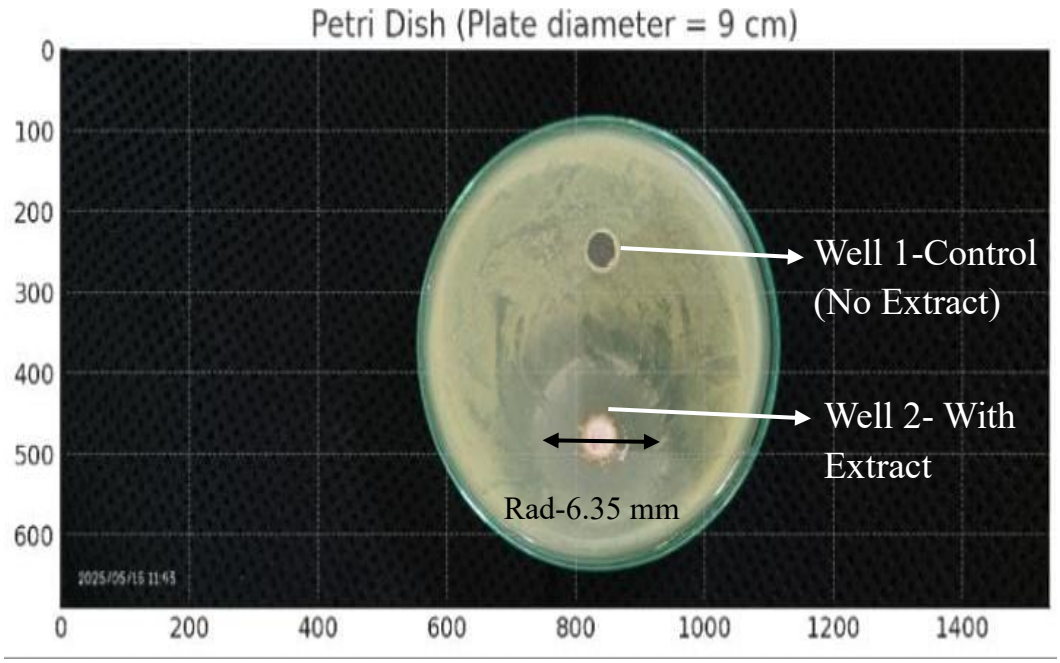


Fig 4.12 ZONE OF INHIBITION

Zone of inhibition was formed in well 2 only. It confirms that the onion peel extract inhibits the biofilm formation.

4.9 BIOFILM INHITION ASSAY USING CRYSTAL VIOLET

4.9.1 QUALITATIVE TEST USING CRYSTAL VIOLET (TEST TUBE METHOD)

Bacterial cultures were inoculated into test tubes containing growth medium and various concentrations of test compounds. Tubes were incubated statically at 37°C for 24–48 hours to allow biofilm formation. Post incubation, non-adherent cells were washed off using PBS. Biofilms were stained with 0.1% crystal violet for 15–20 minutes. Excess stain was rinsed off, and the tubes were dried. Biofilm presence was confirmed by observing a violet-stained ring at the air-liquid interface. Reduction in ring intensity in treated samples compared to controls indicated biofilm inhibition.

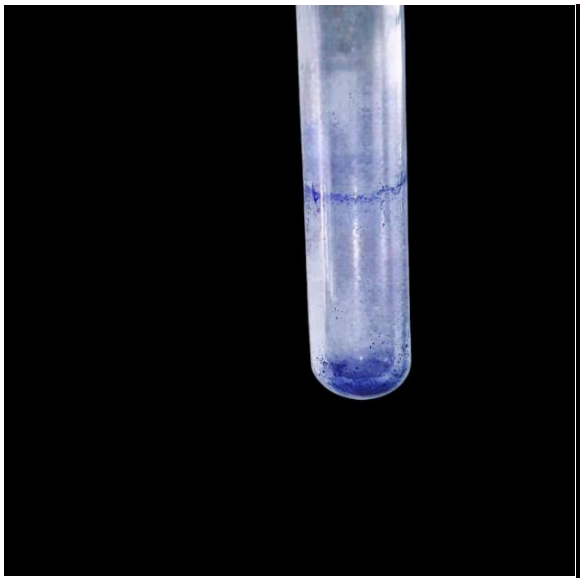


Fig 4.13 Control Tube

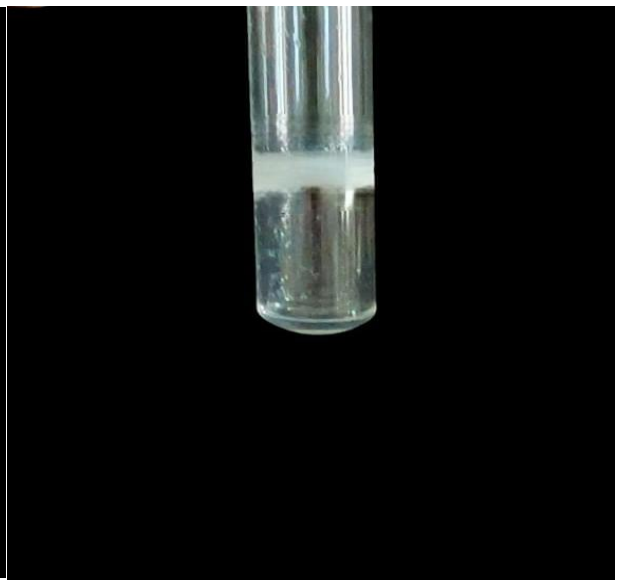


Fig 4.14 Mild Inhibition of Biofilm

Fig 4.15 Moderate Inhibition of Biofilm

Fig 4.16 Strong Inhibition of Biofilm

Table 4.4 Determination of Stain Intensity

TREATMENT	STAIN INTENSITY
Control	Deep Violet ring
25µg/ml	White Pale ring
50µg/ml	Light White ring
100µg/ml	No ring

4.9.2 QUANTITATIVE TEST USING CRYSTAL VIOLET (96 WELL PLATE METHOD)

Biofilm inhibition was assessed using a crystal violet staining assay. Bacterial cultures were incubated in 96-well plates at 37 °C for 24 hours. After removing non-adherent cells and washing, biofilms were heat-fixed at 60 °C for 1 hour. Wells were stained with 0.2% crystal violet for 25 minutes, washed, and air-dried. Bound dye was solubilized with 95% ethanol, and absorbance was measured at 570 nm to quantify biofilm biomass. Inhibition was determined by comparing treated and untreated samples.

Table 4.5 Biofilm Intensity using Crystal Violet

WELL	SAMPLE TYPE	EXTRACT CONC(µg/ml)	OD@570nm (BIOFILM INTENSITY)
A1	Positive control	0	1.0
A2	Positive control	0	0.98
A3	Positive control	0	1.02
B1	25µg/ml	25	0.73

B2	25µg/ml	25	0.7
B3	25µg/ml	25	0.72
C1	50µg/ml	50	0.52
C2	50µg/ml	50	0.5
C3	50µg/ml	50	0.51
D1	75µg/ml	75	0.35
D2	75µg/ml	75	0.33
D3	75µg/ml	75	0.34
E1	100µg/ml	100	0.21
E2	100µg/ml	100	0.18
E3	100µg/ml	100	0.19
F1	Negative control	0	0.05
F2	Negative control	0	0.06
F3	Negative control	0	0.04

Table 4.6 Biofilm inhibition (%)

EXTRAXT CONC(µg/ml)	OD @570nm (Avg)	BIOFILM INHIBITION (%)
25	0.72	28%
50	0.51	49%
75	0.34	66%
100	0.19	81%

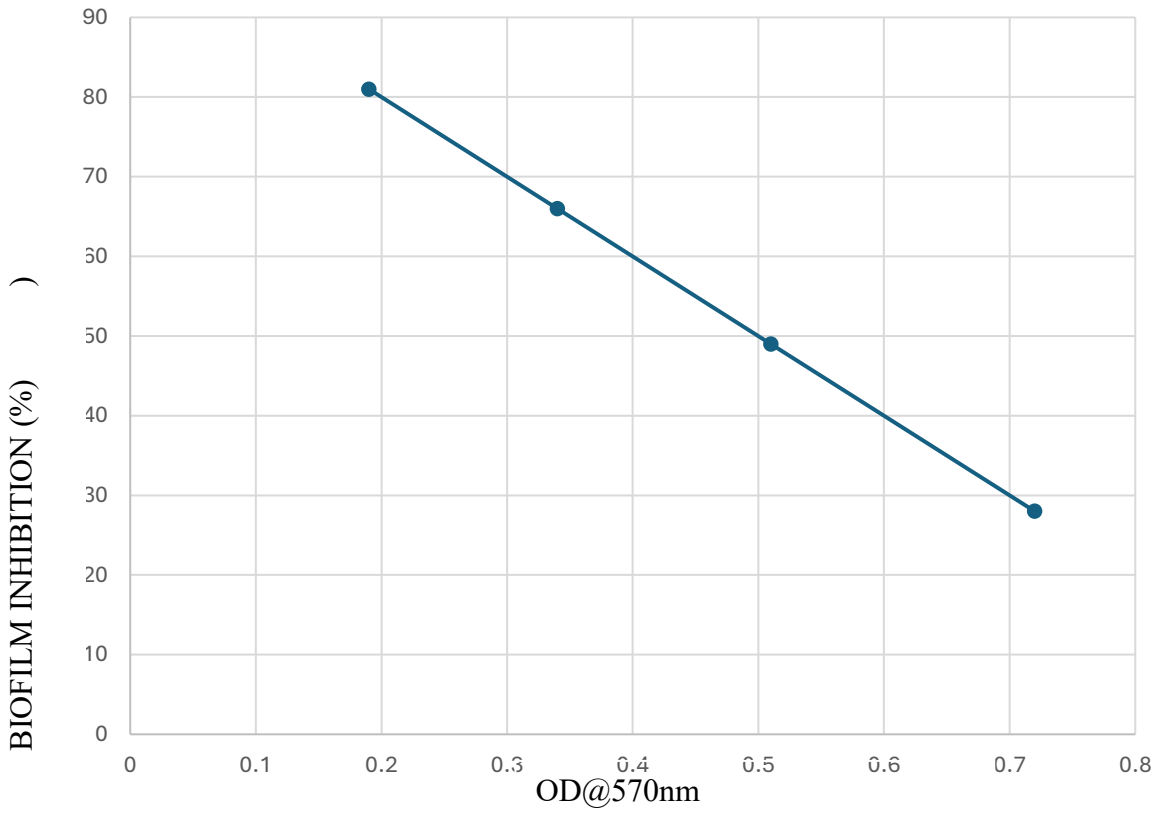


Fig 4.17 BIOFILM INHIBITION (OD)

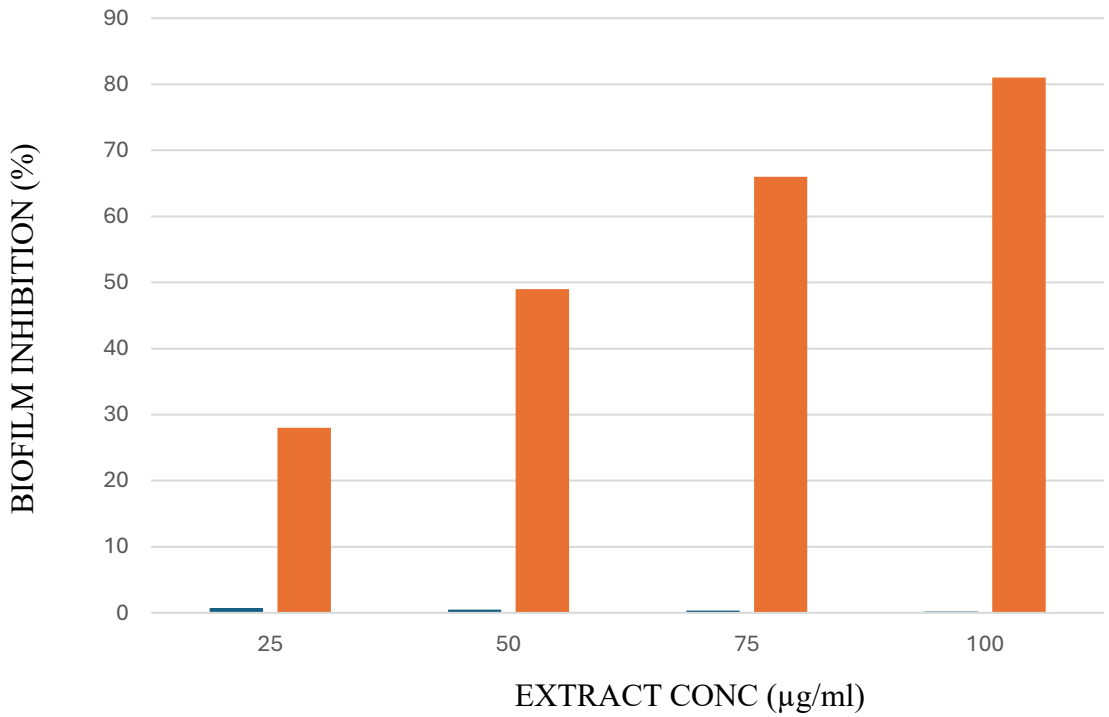


Fig 4.18 BIOFILM INHIBITION (%)

4.10 QUALITATIVE ASSAY USING LIGHT MICROSCOPY

Light microscopy revealed distinct biofilm formation on coverslips, with varying density and structural patterns. Staining confirmed microbial adherence, indicating successful biofilm development under the tested conditions.

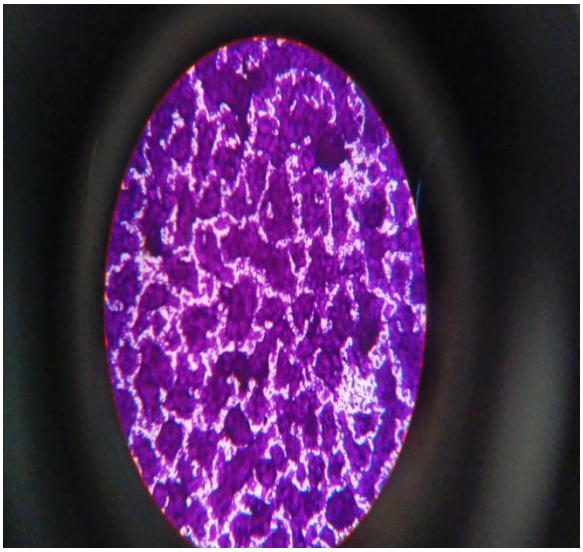


Fig 4.19 Positive Control

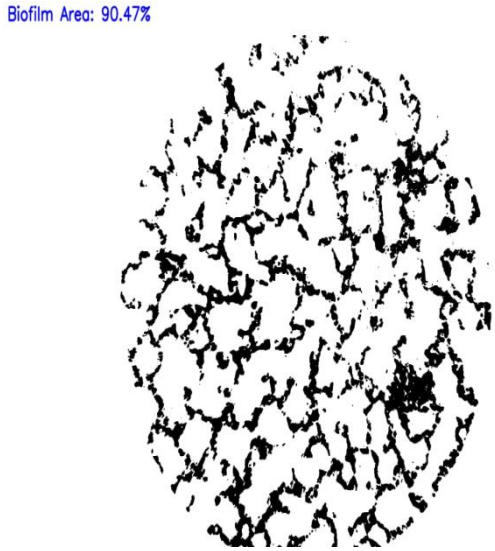


Fig 4.20 Grayscale Image of P.control

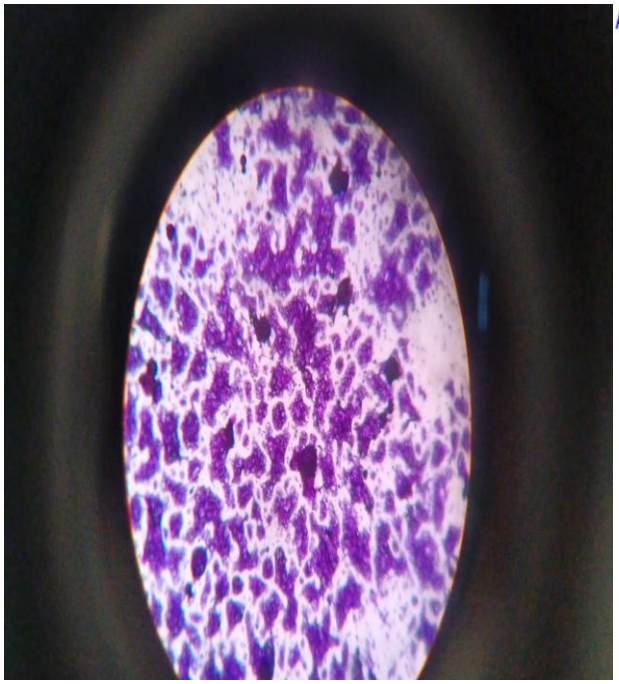
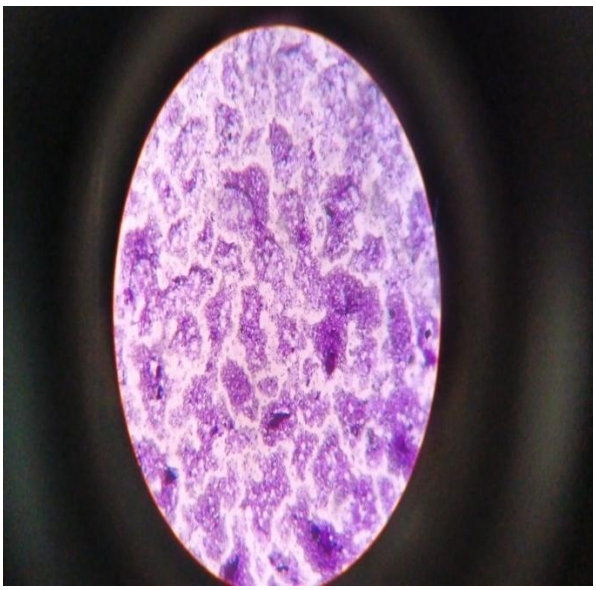


Fig 4.21 Biofilm with Low Conc of Extract

Fig 4.22 Grayscale Image of 4.21



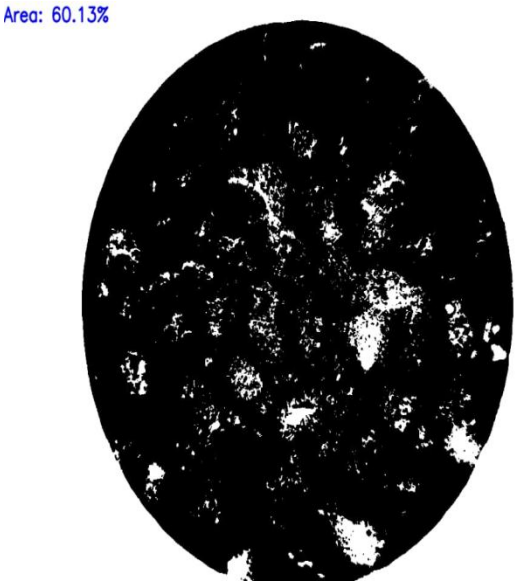


Fig 4.23 Biofilm with High Conc of Extract Fig 4.24 Grayscale Image of 4.23

Positive Control Without onion peel extract with 90.47% of biofilm area. Low Conc of onion peel extract with 71.65% of biofilm area. High Conc of onion peel extract with 60.13% of biofilm area. There is decrease in area of biofilm in high Conc of onion peel extract. Compared to positive control. It confirms the inhibition of biofilm with onion peel extract.

CHAPTER 5

CONCLUSION

This study successfully demonstrates the antibiofilm efficacy of onion peel extract against *Bacillus subtilis*, a known biofilm-forming bacterium. Phytochemical analysis confirmed the presence of active compounds such as flavonoids and phenolics, which are recognized for their antimicrobial and antibiofilm properties. Initial antimicrobial testing using the zone of inhibition method showed clear inhibitory zones, indicating that the extract possesses notable antibacterial activity. Further analysis using light microscopy, the crystal violet microtiter plate assay, and the ring tube confirmatory method revealed a substantial reduction in biofilm formation. Quantitative analysis using the 96-well plate assay showed that onion peel extract inhibited biofilm formation by approximately **60–80%,** depending on the concentration applied. This significant inhibition indicates that the extract effectively interferes with biofilm development, likely by preventing bacterial adherence and disrupting the biofilm matrix. The use of onion peel extract, a low-cost and eco-friendly byproduct, offers a sustainable alternative to synthetic antibiofilm agents. Its ability to inhibit biofilm formation by up to **80%** makes it a promising candidate for applications in healthcare, agriculture, and food industries, where biofilm-associated issues are prevalent. Future studies should focus on isolating the specific bioactive constituents responsible for the observed activity and elucidating their mechanisms of action. In addition, in vivo studies and safety assessments are necessary to validate its potential for clinical or industrial use. Overall, onion peel extract presents an effective natural strategy to combat biofilm-related bacterial persistence and antibiotic resistance.

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